Intelligent Elevator Control By Ordinal Structure Fuzzy Logic Algorithm

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Abstract

With the advancement of intelligent computerised buildings in recent years, there has been strong demands for intelligent elevator control with more sophistication and diverse functions. The design criteria of an intelligent elevator control system would include optimising the movement of a group of elevators with respect to time, energy, load, etc. In this paper, a new elevator group supervisory control system based on the ordinal structure fuzzy logic algorithm is proposed. The system determines the optimum car to answer a hall call using the knowledge and experiential rules of experts. A software has been developed to simulate the traffic flow of three elevator cars in a 15-floor building. The software simulates the movements of the cars as found in practical elevator systems. It can be verified through simulation that the new system can bring about considerable improvements in the average waiting time, riding time, etc. in comparison with conventional methods.

1.0 Introduction

Due to scarcity and high price of land in urban areas, more and more high rise buildings are being constructed. Buildings with thirty floors or more are now a common sight in many cities around the world. In Kuala Lumpur, the Petronas twin towers which are the world's tallest buildings are nearing completion with 88-floors. With such high rise buildings, there is a need for new techniques in building automation. One aspect of building automation is in the development of intelligent elevator systems. In every high rise building, the task of controlling a group of elevators is necessary in order to obtain optimum performance. Two of the main criteria for optimum performance of elevator systems are minimising waiting time and riding time. Other criteria include loading with respect to the number of passengers in the elevator, safety measures, and also comfortability.

In a conventional elevator system, the task of controlling a large number of elevators is numerically evaluated by calculating a specified fixed-evaluation function. It has been realised that knowledge and experiential rules of experts can be incorporated in the elevator system to improve performance [1-4]. However, such expert knowledge is fragmentary and fuzzy which are difficult to organise. Furthermore, the choice of "good" rules and evaluation functions are too complicated in many cases. It is difficult to adequately incorporate such knowledge into products using conventional software and hardware technology.

In order to overcome such problems as described above, a new elevator control system using fuzzy logic algorithm is proposed based on the ordinal structure theory [5,6]. This system determines the optimum car within a group of elevators to answer a hall call using the knowledge and experiential rules of experts. Instead of using the simple up and down hall call buttons, destination oriented keypads at each floor is used. This system requires the passengers to enter their desired floors on the keypad before they enter the car. The system then assigns the passenger the respective optimal car to take through information displayed on dot matrix displays near the keypad. This new elevator supervisory control system has several objectives which can meet users satisfaction. It can improve not only the average waiting time, but also the riding time, load, energy and so on. This paper discusses the design and operations of the proposed fuzzy logic elevator control system. The system is evaluated and tested using an interactive graphic simulator that we developed, simulating the traffic conditions of a three-elevator system in a 15-floor building.

2.0 Elevator Group Control System

The general structure of an elevator group supervisory control system is illustrated as in Fig. 1. In the elevator system, there are two types of passenger requests : hall calls and car calls. A hall call is registered at each hall that represents only the direction to which a passenger at the hall wishes to go, i.e. "up" or "down". A car call is registered in a car and it represents the destination of the passengers in the car.

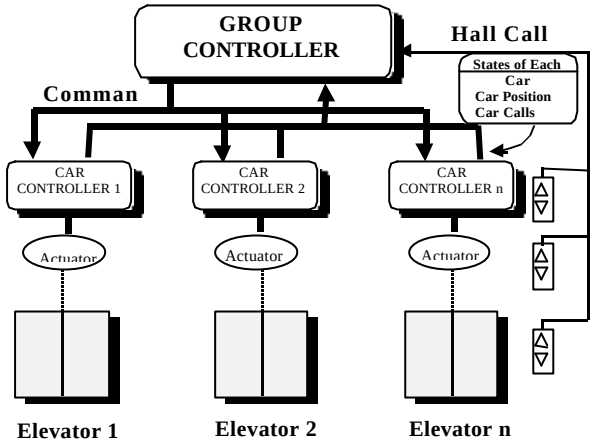


Fig. 1 A general structure of an elevator group supervisory control system.

In the system as given in Fig. 1, there are two types of controllers, one is the car controller and the other is the group controller. A car controller controls the motion of the individual car and transfers information about its states to the group controller. The states of each individual elevator and hall calls from each floor are considered by the group controller. Based on this information the group controller determines the optimum car to answer the hall call which is usually based on a single criteria, i.e., the distance of the elevator from the passenger. Such elevator operation that is based on "up and down hall buttons" and "up and down hall lanterns" has been in used since the beginning of this century. However, this type of system provides some disadvantages such that the supervisory system of the elevators does not receive complete information on the destinations of passengers before they board the car. Consequently, car assignments are based on far less than 50 % of the traffic information that passengers could supply earlier to the system and therefore assignments are obviously poor in quality. The inherent disadvantages of such present conventional system can be improved by using a better elevator technology which is not only destination oriented, but also improves waiting time, comfortabilty, and other factors.

Instead of using the simple up and down hall buttons as in many conventional elevator systems, a keypad similar to that found on touch-tone telephones is used. This keypad can be conveniently located in the elevator lobbies of each floor. This system requires the passengers to enter their desired floors on the keypad before they enter the car. The system then immediately assigns to the passenger the designated car through a display on the keypad.

This system should have the capability to accept boarding of unbooked passenger and also a group of passengers travelling together which tend to register one call only. By calculating of the weight of the load, the unexpected passengers can be detected and the supervisory system can then adjust the actual load accordingly. Similarly, ghost passengers, i.e. passengers whom do not board, can be recognised based on load weighing. The supervisory system can thus cancel such nuisance calls.

3.0 Design Criteria and Constraints

In traditional control system design, there is usually only one performance criteria to be met, such as the optimum speed, position, minimum-time, etc. Due to the limitation of the configuration of the traditional control algorithms, it is difficult to design a system to meet multiple objectives. However, in fuzzy logic control systems, multiple objectives can easily be included in the design. The structure of the fuzzy logic controller makes it possible for a number of important objectives to be met simultaneously. In the design of our elevator group supervisory control system, the following objectives are considered :

- To minimise the waiting time of passengers at a floor
- To minimise the time passengers need to spend in an elevator
- To minimise crowding in an elevator
- To minimise travelling distance of each elevator

However, the objectives mentioned above have several conflicting problems. For example, shortening of waiting time must be sacrificed to some extent for the sake of energy saving and also crowding in the elevators. Thus, the group supervisory control system is required to optimise the multiple design objectives with time and traffic flow constraints.

Similar to many practical elevator systems, the following constraints were assumed in the simulation of the proposed intelligent elevator control system :

i) An elevator will not reverse direction if there is a passenger inside.
ii) The capacity of each elevator is 14 people and when capacity is met, it will bypass hall called floors.
iii) Each elevator travels at a constant speed of 0.5 floor per second.
iv) Serving a floor requires 4 seconds to accomplish. During this time a person may simply walk into or out of the elevator. To serve more than 4 people, 6 seconds is needed.
v) An elevator must not bypass any car call.

4.0 Fuzzy Logic Algorithm

Fuzzy logic is a combination of both numerical and symbolic techniques. It excels in producing exact results from imprecise data and is especially useful in computers and electronic applications. Fuzzy logic differs from classical logic in that statements are no longer black or white, true or false, on or off. In traditional logic an object takes on a value of either zero or one. In fuzzy logic, a statement can assume any real value between 0 and 1, representing the degree to which an element belongs to a given set. The human brain can reason with uncertainties, vagueness and judgements. Computers can manipulate precise valuations. Fuzzy logic is an attempt to combine these two techniques.

The fuzzy reasoning, described as " If X is A then Y is B " is said to be simple and conforms to human language. However in cases where the system has multi inputs and multi outputs, one has to build the fuzzy reasoning rules in multi-dimensional input and output spaces in order to describe the behaviour of the system. This is difficult even for those who know the system well. It is considered that the difficulty is caused by a mismatch between the description of the fuzzy reasoning and the actual image of inference rules which humans have.

Each of the elevators in the elevator group supervisory control system is considered as a multi-input single output (MISO) control system. Using the conventional fuzzy reasoning method the inference rules have to be described in multi dimensional inputs and single output spaces and this is rather difficult to be configured. Thus, an ordinal structure model of fuzzy reasoning which was proposed by Ohnishi et. al. [5,6] is used in this system. The model well coincides with the human image of fuzzy inference rules in cases when the system has many inputs and many outputs. All the fuzzy inference rules are described in one dimensional space for each of the model's input and output. Co-ordination of the rules is done with weights attached to each rule. This model is easier in constructing or modifying fuzzy inference rules than the conventional fuzzy reasoning method as each rule is described in single dimensional space.

For an n-input one-output system, the ordinal structure model is described as follows :

R^i : If X_1 is A_{i1} then Y is B_i

R^j : If X_2 is A_{j2} then Y is B_j

(i, j = 1, 2, ..., n)

where, X_1 and X_2 are the inputs and Y is the output, R^i is the i-th fuzzy rule. A_{i1}, A_{j2}, B_i and B_j are the fuzzy variables and n is the number of rules.

Using the moment method

$$y = \frac{\sum_{i=1}^{n} w_i u_i c_i S_i + \sum_{j=1}^{n} w_j u_j c_j S_j}{\sum_{i=1}^{n} w_i u_i S_i + \sum_{j=1}^{n} w_j u_j S_j} \qquad (1)$$

where, w_i and w_j are the weights of rules R^i and R^j, respectively. μ_i is the truth value of R^i in the premise. C_i and S_i are the central position and the area of the membership function with fuzzy variable B_i, respectively. The structure of this ordinal fuzzy model is shown in Fig. 2.

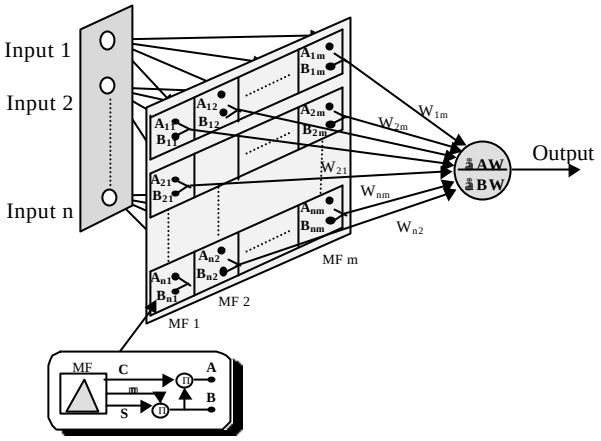


Fig. 2 The structure of the ordinal fuzzy model.

5.0 Properties Of The Elevator Fuzzy Control System

We have developed an elevator fuzzy control system for the supervision and scheduling of 3 elevators with each having a total capacity of 14 persons operating in a 15-floor building. No assumption is made on the arrival patterns of the passengers. However, the passengers are generated randomly accordingly to each floor and operation time. During the morning peak hour, the ground floor will be more crowded than other floors. Thus, the elevator control system has to dispatch all the cars to the crowded floor during peak hours and has a free car parked on the main floor in advanced during off-peak hours, which is considered as the initial stage of the system.

In order to achieve good traffic performance, the elevator fuzzy control system uses six kinds of parameters as the control inputs and one parameter for the output. These parameters represent the criteria or objectives to be optimised in this elevator systems which are as follows:

Waiting Time :-

- Total time an elevator needed to travel from its current position to the new hall call.

Riding Time :-

- Total time a passenger spent in the elevator until he reached as his destination.

Loading :-

- Number of passengers in an elevator.

Travelling Distance :-

- Distance between elevator position and new hall call in terms of number of floors.

Hall call Area Weight :-

- The area weight of the elevator which goes to the floor where a new hall call is generated.

Destination Area Weight :-

- The area weight of the elevator which goes to the floor where the destination of the new hall call is generated.

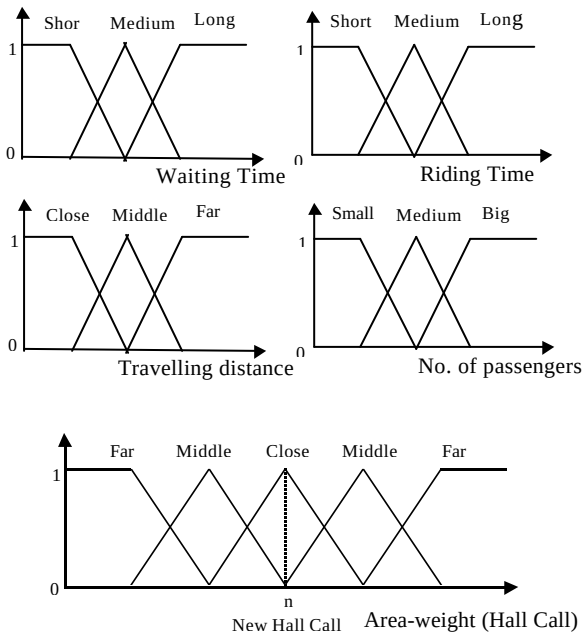
Priority :-

- Output of the fuzzy controller, where the elevator with highest value will be assigned.

As can be observed, it is difficult to configure six kinds of parameters at a time using the conventional fuzzy reasoning method. Thus, the ordinal structure model of fuzzy reasoning is used. With this model, all the fuzzy inference rules are described in one dimensional space for each input and output. The proposed fuzzy inference rules are shown in Table 1 and the membership function of the inputs and output variables are shown in Fig. 3 and Fig. 4.

Table 1. A total of 18 fuzzy inference rules used in the simulation.

Inference Rules	Weight
R^1 : If waiting time is ***Short*** then priority is ***Big***.	0.70
R^2 : If waiting time is ***Medium*** then priority is ***Medium***.	0.70
R^3 : If waiting time is ***Long*** then priority is ***Small***.	
R^4 : If riding time is ***Short*** then priority is ***Big***.	0.40
R^5 : If riding time is ***Medium*** then priority is ***Medium***.	0.40
R^6 : If riding time is ***Long*** then priority is ***Small***.	0.40
R^7 : If loading is ***Small*** then priority is ***Big***.	0.50
R^8 : If loading is ***Medium*** then priority is ***Medium***.	
R^9 : If loading is ***Big*** then priority is ***Small***.	0.60
R^{10}: If travelling distance is ***Close*** then priority is ***Big***.	0.05
R^{11}: If travelling distance is ***Middle*** then priority is ***Medium***.	0.05
R^{12}: If travelling distance is ***Far*** then priority is ***Small***.	0.05
R^{13}: If hall call area weight is ***Close*** then priority is ***Big***.	0.50
R^{14}: If hall call area weight is ***Middle*** then priority is ***Medium***.	
R^{15}: If hall call area weight is ***Far*** then priority is ***Small***.	0.50
R^{16}: If destination area weight is ***Close*** then priority is ***Big***.	0.40
R^{17}: If destination area weight is ***Middle*** then priority is ***Medium***.	0.4
R^{18}: If destination area weight is ***Far*** then priority is ***Small***.	



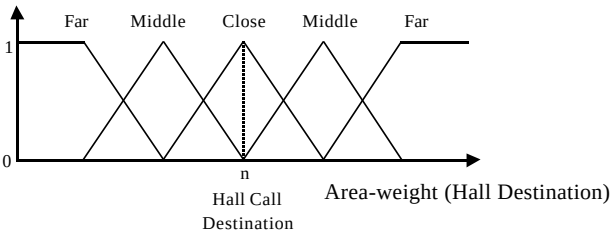


Fig. 3 Membership functions of the elevator system inputs.

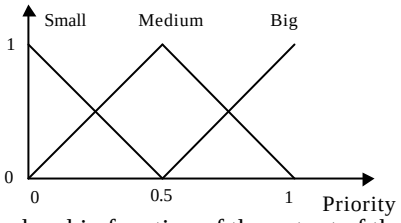


Fig. 4 The membership function of the output of the elevator system.

6.0 Simulation Results and Discussions

To evaluate the feasibility of the proposed fuzzy control system, a highly interactive graphic simulation program is written on an IBM PC compatible system. This simulation simulates the movements of the cars as in a practical elevator system. The software was written in Visual Basic 3.0 using event-driven programming techniques with interactive graphical environment as shown in Fig. 5. This software has a number of facilities for easy manipulation and also analysis of the performance of the fuzzy and conventional elevator systems. Three cases of traffic patterns were experimented and the results of the simulations are given in this section.

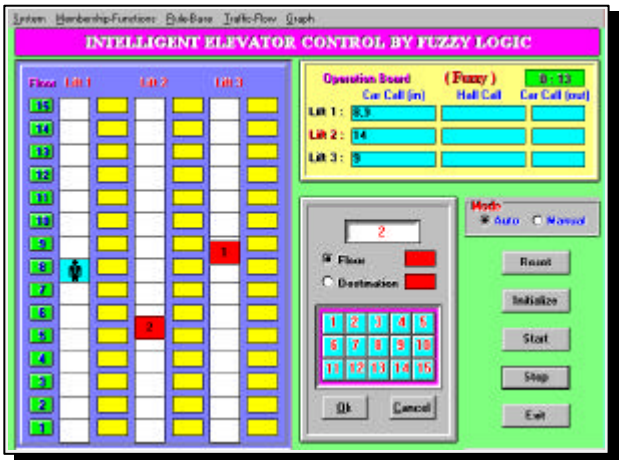


Fig. 5 Simulation of the intelligent elevator control system.

Fig. 6 shows the condition of the 3 elevators for the first case. In this figure the arrow represents the direction of each elevator. The black circle indicates where the car call was initiated for the respective elevators. The black triangle represents a hall call which will be served by the elevator. Finally, a new hall call and its destination generated on the floor are marked by the white triangle and

white circle, respectively. For example, in the same figure, it can be observed that elevator 1 is currently located on the 3rd floor and is moving upwards. It has two car calls on the 8th and 9th floors. A new hall call is then initiated on the 5th floor where its destination is the 14th floor, however no elevator is assigned yet to it.

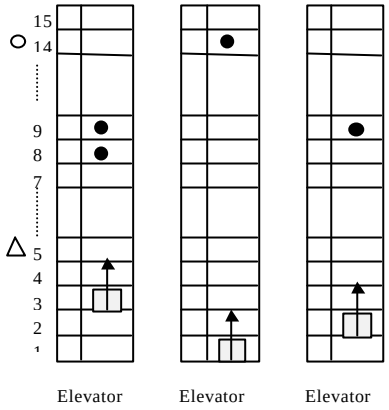


Fig. 6 Initial condition of case 1 simulation.

Using the conventional elevator system which is distance oriented, the elevator closest to the new hall call would, thus, be assigned which in this case is elevator 1. However, the proposed fuzzy system assigned this hall call to the elevator 2. As we find from the initial conditions in Fig. 6, elevator 2 has a car call on the 14th floor. It means that elevator 2 will serve that floor instead of elevator 1. Thus, the destination area weight parameter of elevator 2 will contribute a higher priority in the evaluation function as given by inference rule no.16 in Table 1. In terms of travelling distance, there is a reduction as compared to the conventional system since elevator 1 will finally stop at the 9th floor and not at the 14th floor. Due to this performance, another advantage can be obtained in terms of energy saved.

In the second simulation, all conditions are the same as case 1 except that elevator 1 has three car calls, at 8th, 9th and 14th floor. If a new hall call happens at the 5th floor and its destination is the 14th floor, elevator 1 will thus be selected as the candidate for this assignment. This is because the waiting time of elevator 1 is the shortest and thus contributes a "high" priority to the evaluation function. In this case, the destination area weight parameter gives the same effect to both elevators 1 and 2.

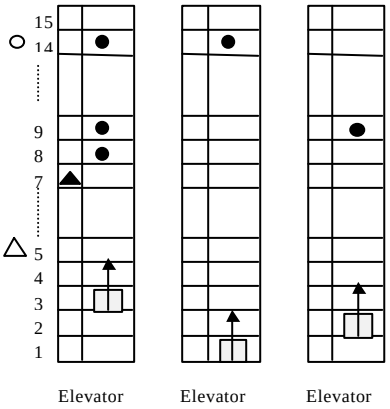


Fig. 7 Initial condition of case 3 simulation.

In the third simulation, a hall call is assigned to elevator 1 at the 7th floor and the rest of the conditions are the same as the previous case. For this case, elevator 2 has been assigned to answer the new hall call instead of elevator 1 even through elevator 1 already has a car call to the 14th floor (see Fig. 7). As we know, elevator 1 will stop three times before it arrives at the 14th floor, thus it is not economical to use this elevator. On the other hand, elevator 2 will move non stop to the 14th floor after picking up the new passenger. Here, it means that the riding time of the passengers is shorter if they travel using elevator 2 compared to elevator 1.

7.0 Conclusion

In this paper, a fuzzy control algorithm for elevator group control has been proposed and evaluated. This new elevator supervisory control system was developed to improve multiple control objectives. The performance of the proposed elevator system is simulated through an interactive graphic simulator that have been developed. It was observed that as compared to conventional system, the proposed elevator system performed better in terms of minimising waiting time, travelling time as well as comfortability in terms of loading. The ordinal structure model of fuzzy logic algorithm provides a simple mechanism for the configuration of rules than conventional fuzzy algorithm.

References

[1] B. K. Chang, S. Kyounga and L. K. Hyung, "A Fuzzy Approach to Elevator Group Control System", IEEE Trans. On System, Man and Cybernetic, Vol. 25 No.4, pp 985-990, 1993.

[2] K. Igarashi, S. Take and T. Ishikawa, "Supervisory Control for Elevator Group with Fuzzy Expert System", Proceeding of the IEEE International Conference on Industrial Technology, pp133-137, 1990.

[3] M. Ho and B. Robertson, "Elevator Group Supervisory Control Using Fuzzy Logic", Canadian Conference on Electrical and Computer Engineering, Vol. 2, pp 825-828, 1994.

[4] S. Tsuji, M. Amano and S. Hikita, "Application of the Expert System to Elevator Group Supervisory Control", The Fifth Conference on Artificial Intelligent Application, pp 287-294, 1991.

[5] Y. Naitoh, T. Furuhashi and Y. Uchikawa, "A Variable Ordinal Structure Model for Fuzzy Reasoning and Its Application to Decision Problem of Working Order", International Conference On Industrial Electronic, Control and Instrumentation, Vol. 2, pp 1539-1543, 1991.

[6] T.Ohnishi, "Fuzzy Reasoning by Ordinal Structure Model of Control Rule", Journal of Japan Society for Fuzzy Theory and Systems, Vol.2, No. 4, pp 125-132, 1990.